HMM-6 — independent numerical, uncertainty and theory audit

Date: 29 October 2026 | **Status:** Independent audit; not a publication acceptance certificate.

Inputs: 4,000 distinct paths, 8 DGPs, 6,000 observations per path (1,200 OOS). **HMM fits:** 4,000 recorded; **models:** 19 specifications and 152 scenario-model aggregates. The full blind-model path-level file was not included in these uploaded aggregates; use the companion Windows audit on the original study directory.

1. Recovery and empirical identification

DGP	True rho	Mean rho	CI low	CI high	State acc.	Majority acc.	nu0 >=99
calibration_anchor	0.9000	0.8999	0.8990	0.9007	0.9040	0.6000	0
full_hmm6	0.8300	0.8277	0.8252	0.8299	0.8843	0.8235	5
iid_mixture	0.0000	0.1862	0.1421	0.2333	0.6085	0.8200	81
scale_only	0.7850	0.7841	0.7813	0.7869	0.8887	0.8372	0
single_student_t	nan	nan	nan	nan	nan	nan	33
tail_only	0.8250	0.3212	0.2720	0.3658	0.6081	0.8571	66
weak_persistence	0.1900	0.1564	0.1060	0.2055	0.5409	0.6543	37
weak_tail	0.8250	0.1052	0.0560	0.1545	0.4517	0.8571	73

State-classification accuracy uses oracle label alignment **only ex post** for evaluation. The constant-state classifier assumes known true stationary state probabilities; it is an easy benchmark, not an operational forecasting algorithm. The bootstrap intervals describe Monte Carlo uncertainty of the *mean estimate across simulated paths*, not per-dataset confidence-interval coverage and not finite-T confidence intervals for a single fitted parameter.

Recovery conclusions

- Calibration-anchor, full-HMM and scale-only DGPs show close recovery of transition persistence; they do **not** justify universal tail parameter identification.
- Tail-only and weak-tail DGPs have severe transition/tail recovery problems even at T=6,000; a simple majority-state label benchmark exceeds the recorded classifier accuracy.
- IID-mixture data have true zero persistence but a positive average fitted rho; this is **false persistence**, not evidence of true hidden-state dynamics.
- The single-Student-t DGP has observationally identical emissions: transitions and regime identity are unidentifiable. Do not report their nominal recovery, standard tests or spurious rho as a conventional identifiable estimand.

2. Numerical implementation audit

- HMM optimizer `success=True`: 4000/4000. One or more fits still below the likelihood at the true parameter vector: **1**. IDs: tail_only__rep0192.
- These are finite local maximization quality checks; maximization over all admissible parameters must be no worse than evaluating the truth.
- **EGARCH numerical penalty contamination:** The custom implementation assigns a `-1e8` per-observation penalty after an invalid conditional-variance recursion. The scenario aggregates listed below have extreme numerical penalties, as revealed by their mean scores; they are **invalid predictive comparisons** without path-level repair.

scenario	model	mean_logscore
iid_mixture	EGARCH11-t	-51668.079

scenario	model	mean_logscore
single_student_t	EGARCH11-t	-130668.073
weak_persistence	EGARCH11-t	-30834.744
weak_tail	EGARCH11-t	-233501.416

- Penalties are implementation artifacts, not evidence of economically catastrophic volatility forecasts. Use the raw per-path file to flag/re-estimate the affected EGARCH paths; never relabel the penalty as a genuine log density.
- Parameter estimates can sit at bounds with apparent numerical convergence; report upper-bound frequencies and restart stability.

3. Predictive-density comparison

scenario	model	hmm_minus_comp_mean_score	SE_difference_upper_bound_Cauchy	lower_95_using_SE_upper_bound	upper_95_using_SE_upper_bound
calibration_anchor	APARCH11-power-symmetric-t	0.02834	0.00887	0.01095	0.04572
calibration_anchor	GARCH11-t	0.03527	0.00902	0.01759	0.05296
calibration_anchor	Static-t	0.13346	0.00904	0.11575	0.15117
full_hmm6	APARCH11-power-symmetric-t	0.00577	0.00394	-0.00195	0.01348
full_hmm6	GARCH11-t	0.00575	0.00394	-0.00197	0.01348
full_hmm6	Static-t	0.02264	0.00408	0.01464	0.03063
iid_mixture	APARCH11-power-symmetric-t	0.00003	0.00220	-0.00429	0.00435
iid_mixture	GARCH11-t	-0.00001	0.00220	-0.00433	0.00431
iid_mixture	Static-t	-0.00010	0.00220	-0.00442	0.00421
scale_only	APARCH11-power-symmetric-t	0.00437	0.00391	-0.00330	0.01204
scale_only	GARCH11-t	0.00440	0.00392	-0.00328	0.01207
scale_only	Static-t	0.02131	0.00406	0.01336	0.02926
single_student_t	APARCH11-power-symmetric-t	-0.00022	0.00217	-0.00447	0.00403
single_student_t	GARCH11-t	-0.00028	0.00217	-0.00453	0.00397
single_student_t	Static-t	-0.00035	0.00217	-0.00460	0.00390
tail_only	APARCH11-power-symmetric-t	0.00002	0.00200	-0.00389	0.00393
tail_only	GARCH11-t	-0.00000	0.00200	-0.00392	0.00391
tail_only	Static-t	-0.00010	0.00200	-0.00402	0.00381
weak_persistence	APARCH11-power-symmetric-t	0.00001	0.00215	-0.00420	0.00421
weak_persistence	GARCH11-t	-0.00003	0.00214	-0.00424	0.00417
weak_persistence	Static-t	-0.00007	0.00214	-0.00428	0.00413
weak_tail	APARCH11-power-symmetric-t	-0.00013	0.00209	-0.00422	0.00396
weak_tail	GARCH11-t	-0.00016	0.00209	-0.00425	0.00394
weak_tail	Static-t	-0.00025	0.00209	-0.00434	0.00385

Difference = HMM-6 minus competitor; positive favors HMM-6. The displayed uncertainty is a **Cauchy upper bound** on the paired standard error computed from marginal MC SEs, not an exact paired CI. It is deliberately conservative, and the 95% endpoints assume approximate normality. Do not report these as final paired hypothesis tests. All paths are matched; calculate the actual 500 path-wise differences using `ALL_COMPETITOR_SUMMARIES.csv`.

4. APARCH reverse attribution

scenario	model	mean_delta	mc_se_delta	mean_gamma	mc_se_gamma
calibration_anchor	APARCH11-power-symmetric-t	0.9991	0.0063	0.0000	0.0000
calibration_anchor	APARCH11-t	1.0011	0.0064	0.0012	0.0020
full_hmm6	APARCH11-power-symmetric-t	1.7112	0.0157	0.0000	0.0000
full_hmm6	APARCH11-t	1.7211	0.0160	0.0025	0.0031
iid_mixture	APARCH11-power-symmetric-t	2.0416	0.0375	0.0000	0.0000
iid_mixture	APARCH11-t	1.9640	0.0394	0.0202	0.0288
scale_only	APARCH11-power-symmetric-t	1.7011	0.0149	0.0000	0.0000
scale_only	APARCH11-t	1.7117	0.0152	0.0051	0.0031
single_student_t	APARCH11-power-symmetric-t	1.9811	0.0313	0.0000	0.0000
single_student_t	APARCH11-t	1.8726	0.0354	0.0397	0.0257
tail_only	APARCH11-power-symmetric-t	1.9404	0.0402	0.0000	0.0000
tail_only	APARCH11-t	1.9234	0.0418	0.0022	0.0312
weak_persistence	APARCH11-power-symmetric-t	1.9858	0.0353	0.0000	0.0000
weak_persistence	APARCH11-t	1.9336	0.0373	0.0221	0.0264
weak_tail	APARCH11-power-symmetric-t	1.9830	0.0359	0.0000	0.0000
weak_tail	APARCH11-t	1.9221	0.0379	0.0632	0.0284

In the strong calibration DGP, estimated delta is close to one; in tail-only DGP, its mean is close to two, while leverage estimates have high cross-replication dispersion. A mean leverage near zero is not proof of correct coverage or zero effects in every path.

5. Theoretical reconciliation — manuscript-ready boundaries

- **Target baseline:** six-parameter centered two-state Student-t HMM, psi=(p00,p11,nu0,nu1,s0,s1). A state-specific mean model adds two parameters and is not HMM-6. With nu_j>2, q_j=(nu_j+3)/(nu_j+1) lies in (1,5/3), **not** (1,3). **Valid local result, for fixed finite T:** under interior/local identification, score regularity, positive observed Fisher information I_T and a locally unique pseudo-true branch, the derivative of psi*_T(epsilon) at epsilon=0+ is I_T^{-1} E_0[S_T W_T]; W_T is constructed with full-sequence *smoothed* state weights and a fixed Gamma shock direction (k,theta).
- **Do not claim:** global Gamma jump identification, a universal sign reversal in fitted q, standard chi-square Sup-LR law at epsilon=0, automatic T-to-infinity convergence, or 6-parameter recovery when the two emissions are identical.
- **Critical separation:** the 4,000-path experiment evaluates the **uncontaminated** HMM-6 estimator and its blind competitors. It does NOT validate the omitted-Bernoulli-Gamma derivative theorem; that requires the separate finite-window impulse/derivative experiment and its own numerical error assessment.
- **Data caveats:** synthetic return-level results do not demonstrate any real-market historical Gamma jumps, risk-neutral pricing or economic welfare improvement. The empirical gap-aware HMM may have eight or more free parameters and must not be labeled a six-parameter model.
- **Manuscript reconciliation:** the earlier 26-Oct proposal PDF describes an eight-parameter mean-switching baseline and asserts global jump identification, standard boundary inference and a generic q sign-reversal. Those assertions are not supported by these data. The later GT_HMM6_Paper.pdf contains the narrower six-parameter finite-window theorem and is the defensible starting point, subject to checking its proofs and independent numerical evidence.
- **Publication blocker:** verify EGARCH implementation; compute exact paired uncertainty; document tail-only weak identification and false persistence, and decide whether to report a conditional identification region rather than a universal applicability claim.

6. Submission-ready wording (suggested)

> In 4,000 independent length-6,000 synthetic samples across eight generating regimes, the centered two-state Student-t HMM recovered transition persistence accurately when regimes were separated by conditional scale. Recovery deteriorated markedly when hidden regimes differed chiefly in tail shape, and an unrestricted HMM inferred positive average persistence under an IID-mixture control. These findings qualify the practical scope of latent-tail identification. The reverse-attribution study covered 19 conventional model specifications; numerical penalty contamination in selected EGARCH recursions requires repair before final comparative inference. The finite-window omitted-shock score sensitivity theorem is a separate local result and is not inferred from these no-jump Monte Carlo experiments.

7. Remaining tasks with no new global Monte Carlo requirement

1. Run the exact paired audit on the original `FULL_T6000_R500/blind_competitor_results/ALL_COMPETITOR_SUMMARIES.csv`; retain all failures, do not let penalty constants enter density scores.
2. Independently verify EGARCH recursion and Student-t likelihood against a trusted reference implementation; selectively rerun **only** affected numerical cases (if needed).
3. Check all six-parameter HMM theoretical assumptions and observational identity / label conventions. Report finite-T theorem separately from global or long-run conjectures.
4. Consolidate the validated DGP tables with the existing finite-window Gamma-shock numerical experiment. Include negative and null outcomes explicitly.
5. Freeze experiment scripts, seeds, dates, versions, and input/output hashes in the public reproducibility archive.